CSE400: Fundamentals of Probability in Computing
Lecture 20: Linear Transformations and Expectations of Multiple RVs
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1. Lecture Outline

Statistical Representation of Random Vectors
Mean Vector, Correlation, and Covariance Matrices
Linear Transformations
Mapping properties under $\underline{Y} = \underline{AX} + \underline{b}$
Multivariate Gaussian Distributions
From 1D/2D cases to N-dimensional Joint PDFs
Special Cases and Properties
Implications of zero correlation and matrix factorization

2. Statistical Representation of Random Vectors

2.1 Random Vector Definition

For random variables $X_1, X_2, \ldots, X_N$, the representation as a vector is called a random vector, denoted as $\underline{X}$.
It provides a compact representation of multiple random variables.
Vector form: $\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$.

2.2 Mean Vector

The mean vector of $\underline{X}$ is defined as $\underline{\mu}_X = E[\underline{X}]$.
It represents the average value of the random vector.
Each element corresponds to the mean of one random variable.
Example structure: $\underline{\mu}_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$.

2.3 Correlation Matrix

The correlation matrix represents the correlation between multiple random variables, defined as $R_{XX} = E[\underline{XX}^T]$.

Matrix elements are defined as $R_{XX}(i,j) = E[X_i X_j]$, where i and $j \leq N$.
R_{XX} is a symmetric matrix.

Full matrix representation: $R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \dots & E[X_1X_N] \\ E[X_2X_1] & \dots & \dots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \dots & E[X_NX_N] \end{bmatrix}_{N \times N}$.

2.4 Covariance Matrix

The covariance matrix acts as a relationship map of random vectors.
Mathematical definition for a random vector: $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.

Practical structure: $C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \dots \\ Cov(X_2, X_1) & Var(X_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$.

Symmetry property: $Cov(X_i, X_j) = Cov(X_j, X_i)$, which makes the matrix a "mirror" of itself.

3. Linear Transformations

3.1 The Linear Model

Consider the transformation: $\underline{Y} = \underline{AX} + \underline{b}$.
$\underline{A}$ represents the transformation matrix, which scales and rotates the data.
$\underline{b}$ represents the constant vector, which shifts or translates the center.
$\underline{X}$ represents the input random vector.
For N-dimensional vectors, the system of equations is:
$Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1$
$Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N$

3.2 Summary of Properties under Transformation

If $\underline{Y} = \underline{AX} + \underline{b}$, the resulting statistical properties are summarized below:

Property	**Transformed Result**
Mean	$\mu_Y = \underline{A}\mu_X + \underline{b}$
Covariance	$C_{YY} = \underline{A}C_{XX}\underline{A}^T$
Correlation	$R_{YY} = \underline{A}R_{XX}\underline{A}^T + \underline{A}\mu_X\underline{b}^T + \underline{b}\mu_X^T\underline{A}^T + \underline{bb}^T$

4. Derivations

4.1 Mean Transformation Derivation

Definition: $\mu_Y = E[\underline{Y}]$.
Substitute the linear model: $\mu_Y = E[\underline{AX} + \underline{b}]$.
Apply linearity of expectation: $E[\underline{AX}] + E[\underline{b}]$.
Final Result: $\mu_Y = \underline{A}\mu_X + \underline{b}$.

4.2 Correlation Matrix Transformation Derivation

Definition: $R_{YY} = E[\underline{YY}^T] = E[(\underline{AX} + \underline{b})(\underline{AX} + \underline{b})^T]$.
Expansion of terms: $R_{YY} = E[\underline{AXX}^T\underline{A}^T + \underline{AXb}^T + \underline{bX}^T\underline{A}^T + \underline{bb}^T]$.
Applying Expectation yields the final result: $R_{YY} = \underline{A}R_{XX}\underline{A}^T + \underline{A}\mu_X\underline{b}^T + \underline{b}\mu_X^T\underline{A}^T + \underline{bb}^T$.

4.3 Covariance Matrix Transformation Derivation

Key Observation: The deviation from the mean is independent of the shift $\underline{b}$.
$\underline{Y} - \mu_Y = (\underline{AX} + \underline{b}) - (\underline{A}\mu_X + \underline{b}) = \underline{A}(\underline{X} - \mu_X)$.
Derivation steps:
$C_{YY} = E[(\underline{Y} - \mu_Y)(\underline{Y} - \mu_Y)^T]$.
$C_{YY} = E[\underline{A}(\underline{X} - \mu_X)(\underline{X} - \mu_X)^T\underline{A}^T]$.
Simplified Result: $C_{YY} = \underline{A}C_{XX}\underline{A}^T$.
Notice that the shift $\underline{b}$ does not affect the covariance, meaning it does not affect the spread.

5. Gaussian Distributions

5.1 1D and 2D Gaussian Distributions

Univariate Gaussian (1D): For a single random variable X, the probability density function (PDF) is $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}exp(-\frac{(x-\mu)^2}{2\sigma^2})$.
Joint Gaussian (2D): For two random variables X and Y, the joint PDF is $f_{XY}(x,y) = \frac{exp\{-\frac{1}{2(1-\rho_{XY}^2)}[(\frac{x-\mu_X}{\sigma_X})^2 - 2\rho_{XY}\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + (\frac{y-\mu_Y}{\sigma_Y})^2]\}}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}}$.

5.2 Multivariate Gaussian Distribution

For an N-dimensional random vector $\underline{X} = [X_1, X_2, \ldots, X_N]^T$.
The General Vector PDF is defined as: $f_{\underline{X}}(x) = \frac{1}{\sqrt{(2\pi)^N det(C_{XX})}}exp(-\frac{1}{2}(x - \underline{\mu}_X)^T C_{XX}^{-1}(x - \underline{\mu}_X))$.
Structure Components: $\underline{\mu}_X$ is the Mean vector of size $(N \times 1)$, and C_{XX} is the Covariance matrix of size $(N \times N)$.

5.3 Special Case: Uncorrelated Gaussian Random Variables

For mutually uncorrelated Gaussian random variables, $Cov(X_i, X_j) = 0$ for all $i \neq j$.
The covariance matrix simplifies to a diagonal matrix, which implies the inverse C_{XX}^{-1} is also a diagonal matrix containing elements σ_n^{-2} on the diagonal.
Substituting the inverse back into the density function yields the exponent: $exp(-\frac{1}{2}\sum_{n=1}^{N}(\frac{x_n-\mu_n}{\sigma_n})^2)$.
Interpretation: The joint Gaussian PDF factorizes into the product of individual Gaussian PDFs: $f_{\underline{x}}(\underline{x}) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} exp(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2})$.

*6. Worked Examples

6.1 Numerical Linear Transformation

Given the linear transformation $\underline{Y} = \underline{AX}$ where:
$\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$, and $R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$.

Find Mean Vector of Y:
$\underline{\mu}_Y = \underline{A}\underline{\mu}_X = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$.

Find Correlation Matrix of Y:
Step 1: Compute $\underline{A}R_{XX} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$.
Step 2: Multiply by $\underline{A}^T$ to get $R_{YY} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$.

Find Covariance Matrix of Y:
Use property $C_{YY} = R_{YY} - \underline{\mu}_Y \underline{\mu}_Y^T$.
Outer product of the mean: $\underline{\mu}_Y \underline{\mu}_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$.
Subtraction: $C_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$. (Note: Source 283 shows an intermediate matrix with -9 in corners, corrected to -11 based on R_{YY}).

6.2 2D Joint Gaussian PDF Derivation

Given mean vector $\underline{\mu}_X = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ and covariance matrix $C_{XX} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$.
Determinant computation: $det(C_{XX}) = \sigma_1^2\sigma_2^2 - (\rho\sigma_1\sigma_2)^2 = \sigma_1^2\sigma_2^2(1-\rho^2)$.
Inverse of covariance matrix: $C_{XX}^{-1} = \frac{1}{det(C_{XX})} adj(C_{XX}) = \frac{1}{1-\rho^2} \begin{bmatrix} \sigma_1^{-2} & -\rho\sigma_1^{-1}\sigma_2^{-1} \\ -\rho\sigma_1^{-1}\sigma_2^{-1} & \sigma_2^{-2} \end{bmatrix}$.
Evaluate quadratic form $Q(x) = (\underline{x} - \underline{\mu}_X)^T C_{XX}^{-1} (\underline{x} - \underline{\mu}_X)$.

After algebraic simplification, this yields: $Q(x) = \frac{1}{1-\rho^2}[(\frac{x_1-\mu_1}{\sigma_1})^2 - 2\rho(\frac{x_1-\mu_1}{\sigma_1})(\frac{x_2-\mu_2}{\sigma_2}) + (\frac{x_2-\mu_2}{\sigma_2})^2]$.

7. Case Study: Smart City Traffic + Weather System (Ahmedabad)

7.1 System Definition

Design of an intelligent system predicting traffic congestion using transportation and climate variables.
Define a random vector $\underline{X}$ of size $N = 5$:
X_1: Traffic flow (vehicles/min), mean = 120.
X_2: Average vehicle speed (km/h), mean = 35.
X_3: Rainfall intensity (mm/hr), mean = 2.
X_4: Temperature (°C), mean = 32.
X_5: Air pollution index (AQI), mean = 180.

7.2 Dependencies (Covariance Map)

The covariance matrix acts as the dependency map of the city. By definition and intuition:
$Cov(X_1, X_3) > 0$: More rain implies more congestion (traffic increases).
$Cov(X_2, X_3) < 0$: More rain implies lower speed.
$Cov(X_1, X_5) > 0$: More traffic implies higher pollution.
$Cov(X_4, X_5) < 0$: Higher temperature may cause pollution to disperse.

7.3 Probabilistic Modeling

Assume the entire system follows a Joint Gaussian Distribution: $\underline{X} \sim \mathcal{N}(\mu, \Sigma)$.
Gaussian assumption is used due to the Central Limit Theorem (aggregation of many small effects) and because sensor noise is often Gaussian.
This model allows for easier analytics via closed-form solutions.

7.4 Transformation into Congestion Index

Suppose the city defines a derived metric, the Congestion Index, via linear transformation: $Y = 0.6X_1 - 0.4X_2 + 0.8X_3$.
Since the input is Gaussian, the linear transformation Y is also Gaussian.
The mean of Y is given by $E[Y] = a^T\mu$, and the variance is $Var(Y) = a^T\Sigma a$.

8. Homework Assignments

8.1 Finding the Mean Vector

Problem Statement: Given a random vector $\underline{X}$ with matrices $C_{XX} = \begin{bmatrix} 5 & -1 & 2 \\ -1 & 5 & -2 \\ 2 & -2 & 8 \end{bmatrix}$ and $R_{XX} = \begin{bmatrix} 6 & 1 & 1 \\ 1 & 9 & -4 \\ 1 & -4 & 9 \end{bmatrix}$, find the mean vector $\underline{\mu}_X$.

Hint: Use the relationship $C_{XX} = R_{XX} - \underline{\mu}_X \underline{\mu}_X^T$ and rearrange to solve for the outer product: $\underline{\mu}_X \underline{\mu}_X^T = R_{XX} - C_{XX}$. Identify elements of $\underline{\mu}_X$ from the diagonal of the result.

8.2 Variance Proof

Homework: Prove the following variance property: $Var[X + Y] = Var[X] + Var[Y] + 2Cov(X, Y)$.